

History of Plants biotechnology



Early Plant breeding

- Humans domesticate crops.
- Breed plants to further improve desirable Characteristics .

The word "**biotechnology**" was first used in 1917 to describe processes using living organisms to make a product or run a process, such as industrial fermentation. In fact, biotechnology began when human started to plant their own crops, domesticate an ,ferment juice into wine ,make cheese,leaven bread.

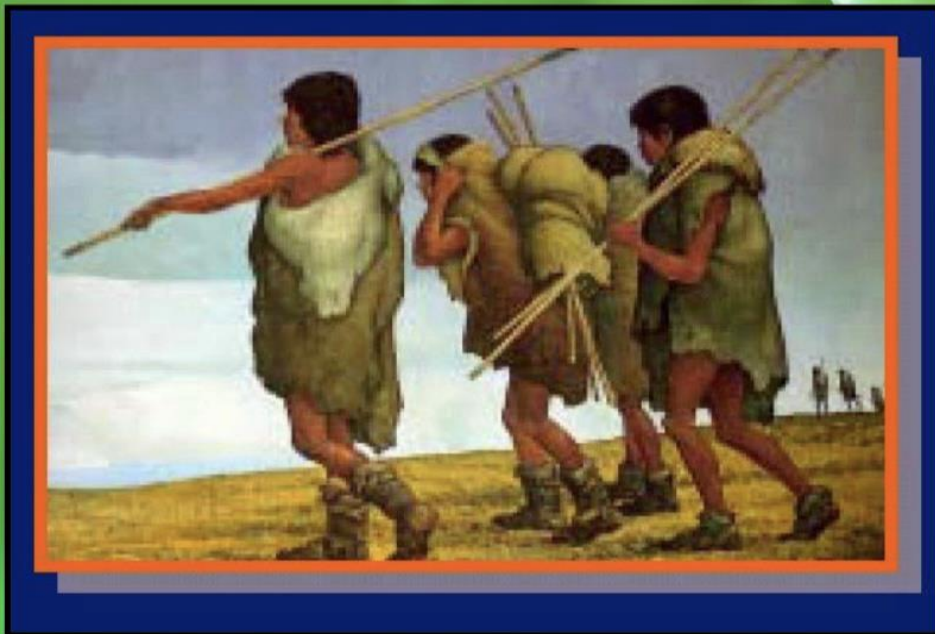
Present definition of biotechnology

“Any technological application that uses biological systems , living organisms, derivatives theory, to make or modify products or processes for specific use”.

Note::-

Make:-new products ,modify:-use some genome of specific organisms (over expression,down regulation,depression).

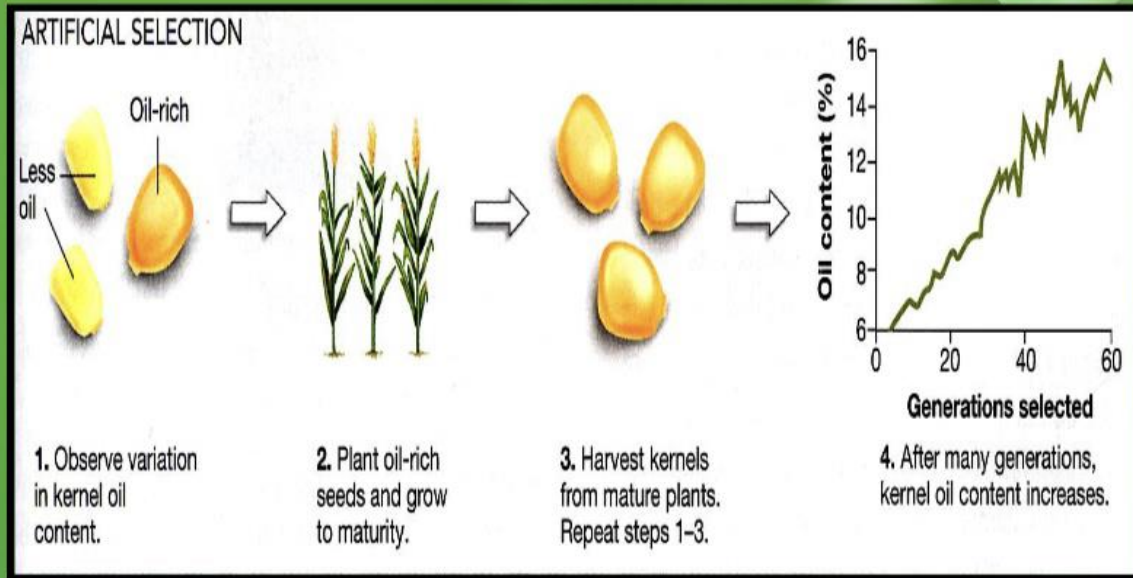
Plant breeding 12,000 years ago ...



Bread-making wheat grown in Egypt, rice cultivated in China



Artificial selection of kernel oil content



Plant Biotechnology

- Plant biotechnology is founded on the principles of cellular totipotency and genetic transformation.
- Traced back to the Cell Theory of M.J.Schleiden and T.Schwann.
- Discovery of the genetic transformation in bacteria by Frederick Griffith.

Plant biotechnology

- “Plant biotechnology describes a precise process in which scientific techniques are used to develop useful and beneficial plants “

Introduction

- Eighteenth Jan 1983, three independent groups described *Agrobacterium tumefaciens*-mediated genetic transformation, leading to the production of normal, fertile transgenic plants.

- Introduced a bacterial antibiotic resistance gene into tobacco.

Objective

- Scientists believe that the new technology would allow introduction of agronomically important genes into crop species that are susceptible to agro-bacterial infections.

****THE CELL THEORY (Schleiden&Schwann).**

Schleiden (1838):

- Every plant and animal is “an aggregate of fully individualized, independent, separate beings”, that is, cell.

Schwann (1839):

- “Some of these elementary parts (cells) which do not differ from others, are capable of being separated from the organism and continuing to grow independently.

Cell Theory:

- 1• All living things are composed of cells.
- 2• Cells are the basic unit of structure and function in living things.
- 3• Cells arise from pre-existing cells.

Virchow (1858):

- “Where a cell arises, there must have been a cell before, even as an animal can come from nothing but an animal, a plant from nothing but a plant...nor can any developed tissue be traced back to anything but a cell”.

Revival of Cell Theory:

- Interest in the Cell Theory was revived in 1858 by the famous aphorism of Ludwig Karl

Virchow:

“Omnis cellula a cellula” (all cells arise from cells).

- He determined that diseased cells always arose

from healthy cells, and rejected the prevailing view of spontaneous generation.

- Herman Vöchting (1878), attempted to demonstrate totipotency experimentally by dissecting and growing smaller and smaller fragments of plant tissues.

Totipotency Gottlieb Haberlandt.

- Totipotency is the ability of a single cell to divide and produce all the differentiated cells in an organism, including extra embryonic tissues.
- Gottlieb Haberlandt was the first to try to obtain experimental evidence of totipotency by culturing plant cells in nutrient solutions in hope of regenerating whole plants.
- He cultured chloroplast containing differentiated cells from leaves of *Lamium purpureum* in Knop's solution in hanging drop culture

Results

- The cells grew in size but did not undergo any cell divisions. eventually the cultures were lost to infection.
- He failed largely because of his unfortunate choice of experimental material, and because of inadequacy of nutrition provided by Knop's salt solution.

Predictions made by Gottlieb Haberlandt

He advocated the use of embryo sac fluids(Coenocytic liquid endosperm, such as coconut milk that was later used in tissue culture).

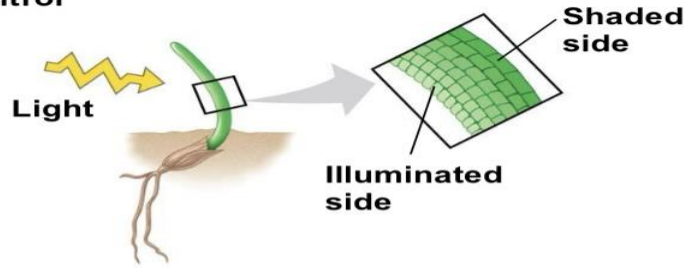
- Inducing cell divisions in vegetative cells.
- Charles Darwin(1880) was one of the first to observe and describe the curvature of seedlings towards light.
- Effect could be prevented by covering the tip of the colioptile.

- The hormone later to be identified as naturally occurring auxin, IAA.

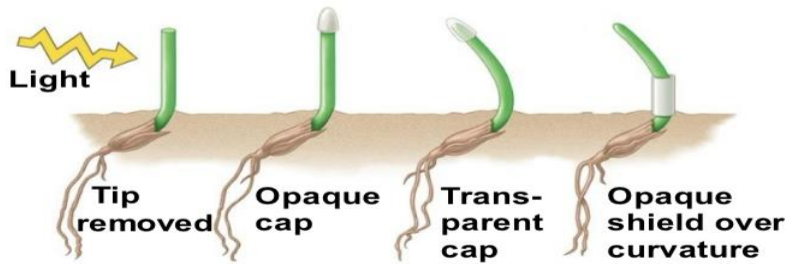
Auxins , • Auxins are a class of plant growth substance (often called phytohormone or plant hormone). Auxins play an essential role in coordination of many growth and behavioral processes in the plant life cycle.

Figure 39.5
Results

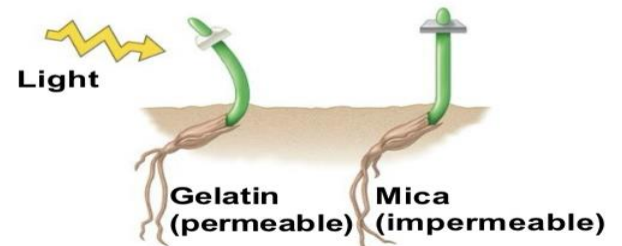
Control



Darwin and Darwin



Boysen-Jensen



Data from C. R. Darwin, *The power of movement in plants*, John Murray, London (1880). P. Boysen-Jensen, *Concerning the performance of phototropic stimuli on the Avenacoleoptile*, *Berichte der Deutschen Botanischen Gesellschaft (Reports of the German Botanical Society)* 31:559–566 (1913).

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*Opaque cap , doesn't bent , Does not allow light to pass through

*Shield over curvature this tip On the bend area.

**Boysen Jensen

Gelatin

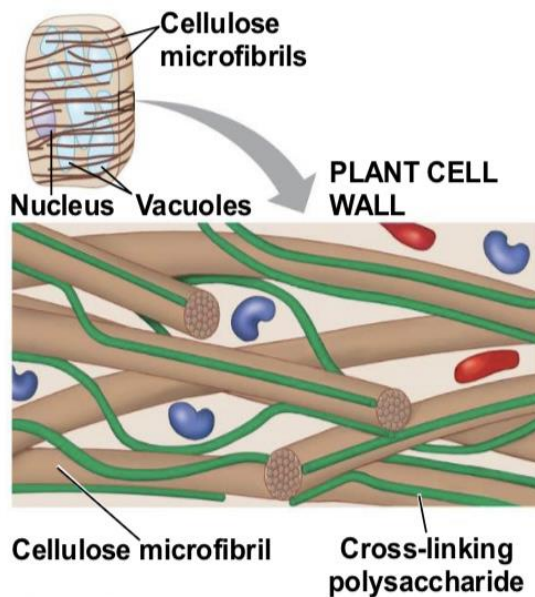
1- Permeable تنحني

2- Mice imperial لا تنحني

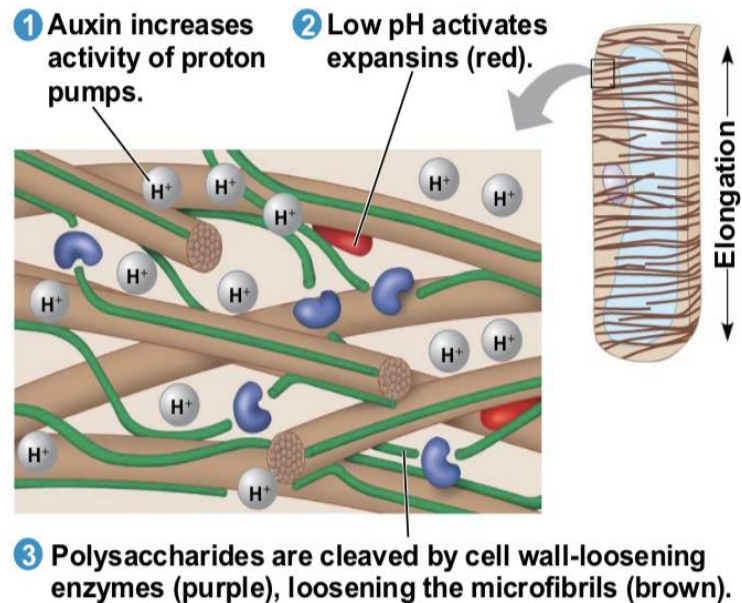
Auxin , difusable mateial one cell to another.

Figure 39.7

Cell wall before auxin activates proton pumps



Loosening of cell wall, enabling elongation



The image showed the mechanism of elongation of plant cells as a result of the effect of the hormone

oxin on the cell wall. The process is divided into two stages:

1. Before Activating Proton Pumps By Oxin:

The cell wall contains cellulose microfibrils and synaptic sugars. cross linking polysaccharides making it rigid (and inflexible .-

There are a number of proteins and enzymes within the cell wall, but they are not yet active.

2. Ease the cell wall and enable elongation

Auxin increases the activity of proton pumps, resulting in the injection of hydrogen ions ($+H$) to the cell wall, reducing the pH (pH).

- (2) Low pH leads to the activation of expansins (indicated red) proteins, which contribute to the dissociation of cellulosic fibers.
- (3) The synaptic sugars are broken down by designated enzymes (indicated as purple), which dilutes the structure of the cell wall, allowing the cellulosic fibers to slide over each other.

As a result of these changes, the cell wall becomes more flexible, allowing the cell to elongate due to internal fullness pressure.

Cytokinins

In addition to auxins another compound is •
cytokinins having profound impact on the
.culture of isolated plant cells
They were discovered by Folk Skoog and his •
colleagues at the university of Wisconsin

Control of Cell Division and Differentiation

Cytokinins are produced in actively growing tissues ▪
such as roots, embryos, and fruits

Cytokinins work together with auxin to control cell ▪
division and differentiation

Anti-aging Effects

Cytokinins slow the aging of some plant organs by ▪
inhibiting
protein breakdown, stimulating RNA and protein
,synthesis
and mobilizing nutrients from surrounding tissues.

Gibberellins have a variety of effects, such as stem ▪
elongation, fruit growth, and seed germination

Stem Elongation

Gibberellins are produced in young roots and leaves ▪

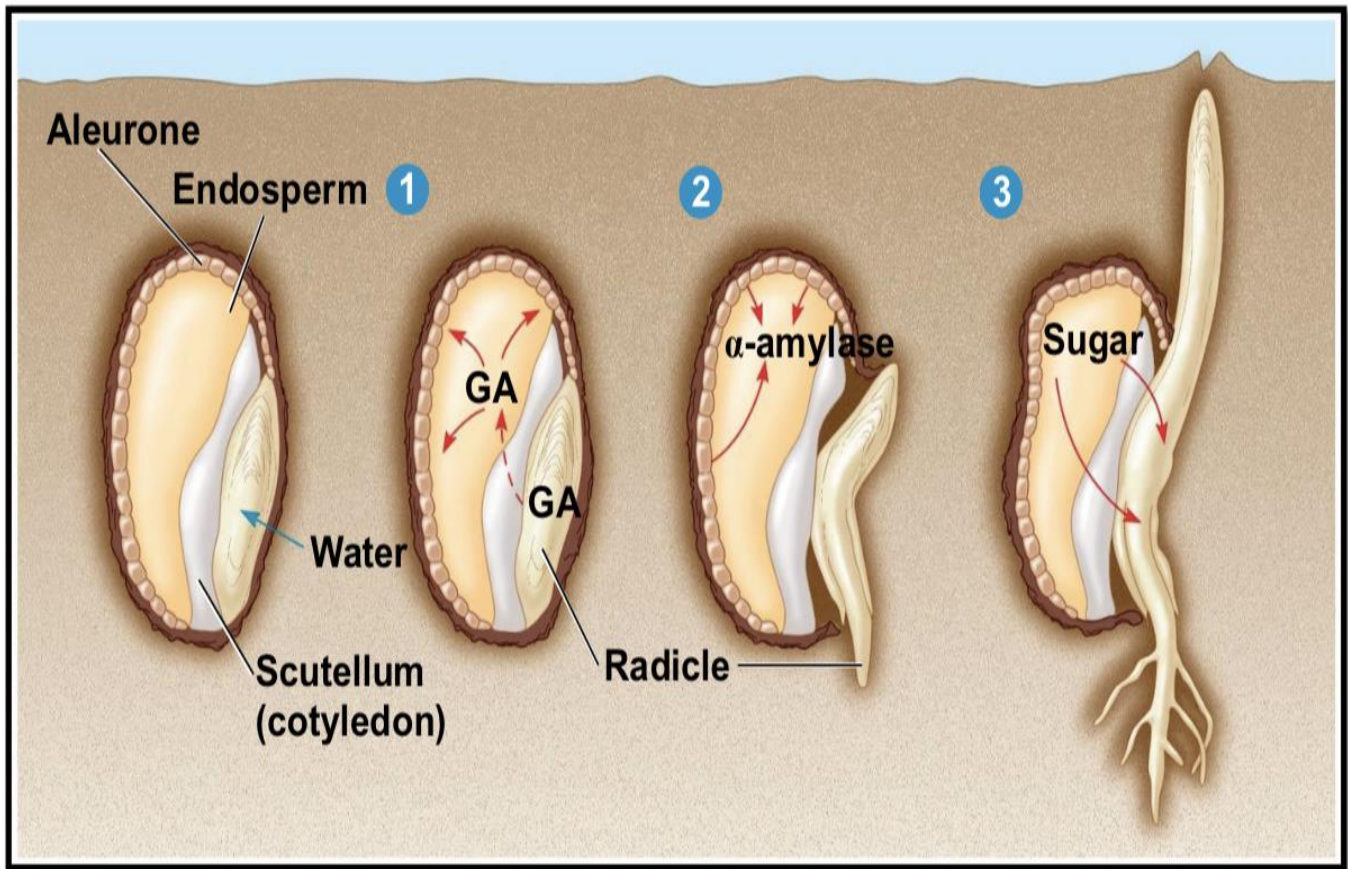
Gibberellins stimulate growth of leaves and stems by ▪
enhancing cell elongation and cell division

Bolting, rapid growth of the floral stalk, is induced by ▪
Gibberellins.



(a) Rosette form (left) and gibberellin-induced bolting (right)





The picture shows the stages of seed germination and the role of the hormone gibberellin (GA - Gibberellic

Acid) in stimulating this process. Germination can be divided into three main stages

1-Water absorption and activation of gibberellin (GA)

- The seed begins to absorb water from the soil, which leads to its swelling and activation of metabolic processes.

- The hormone gibberellin (GA) is secreted from the embryo inside the seed, and spreads to the aleurone layer surrounding the endosperm.

2-Production of the enzyme amylase to break down starch

- Gibberellin stimulates the aleurone layer to produce the enzyme α -amylase, an enzyme that breaks down the starch stored in the endosperm into simple sugars.

- These sugars are provided to the developing embryo to support its growth and development.

- At this stage, the primary root (Radicle) begins to appear outside the seed.

3- Transport of sugars and seedling growth

The decomposed sugars are transported to the developing parts of the embryo, which helps in their expansion and growth.

The seedling begins to appear, and the roots grow to penetrate the soil in search of water and nutrients.

The importance of this process:

These stages are necessary for seed germination and the formation of the new plant, as the embryo initially depends on the food supply inside the seed before it becomes capable of photosynthesis and depends on itself to produce energy.

Absciscic Acid

- Absciscic acid (ABA) slows growth, often by antagonizing the actions of growth hormones
- ABA has many other effects on plants including seed dormancy and drought tolerance

Seed Dormancy

- Seed dormancy increases the likelihood that the seed will germinate only in optimal conditions
- Many dormant seeds germinate when ABA is removed or inactivated
- The ratio of ABA to gibberellins often affects whether seeds will break dormancy
- Precocious (early) germination can be caused by inactive or low levels of ABA

Ethylene

- Plants produce ethylene in response to stresses such as drought, flooding, mechanical pressure, injury, and infection

- The effects of ethylene include response to mechanical stress, senescence, leaf abscission, and fruit ripening.

****The Triple Response to Mechanical Stress**

- Ethylene is produced when a seedling tip pushes against an obstacle
- The production of ethylene induces a triple response in which stem elongation is slowed, the stem thickens, and the stem begins to grow horizontally
- Vertical growth resumes when the effects of the ethylene wear off.

Senescence

- Senescence is the programmed death of certain cells or organs or entire plants
- A burst of ethylene is associated with the onset of apoptosis, programmed cell death

Leaf Abscission▪

A change in the balance of auxin and ethylene controls leaf abscission, the process that occurs in autumn when a leaf falls.

More Recently Discovered Plant Hormones

- Brassinosteroids are chemically similar to cholesterol and the sex hormones of animals
- They induce cell elongation and division in stem segments and seedlings at low concentration
- They slow leaf abscission and promote xylem

Differentiation •

Jasmonates, including jasmonate (JA) and

methyl jasmonate (MeJA) play important roles in plant defense and development

- They are produced in response to wounding and are involved in controlling plant defenses

- Jasmonates also regulate many other physiological processes, including

- Nectar secretion
- Fruit ripening
- Pollen production
- Flowering time
- Seed germination
- Root growth
- Tuber formation
- Mycorrhizal symbiosis
- Tendril coiling

Genetic Transformation

A process by which the genetic material carried by an individual cell is altered by the incorporation of foreign (exogenous) DNA into its genome.

- Genetic Transformation was discovered by chance not by design by Frederick Griffith in

.1928

- He was trying to develop a vaccine for prevention of pneumonia epidemic.

Frederick Griffith Experiment

- Organism used was *Streptococcus pneumoniae*
- He used two different strains of *Streptococcus pneumoniae*
- R strain (Rough type)
- S strain (Smooth type)

Griffith's Experiment

- He injected mice with R strain and S strain.
- The mice injected with S strain develop pneumonia and died after few days, while those infected with R strain remains healthy.
- Mice injected with a mixture of inactivated S strain and normal R strain developed pneumonia. Griffith.

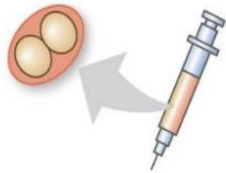
Griffiths Results

- He postulated that some unknown “Transforming Principle” from the inactivated S strain had converted the non-capsulated non-virulent R strain into encapsulated and virulent S strain.

Figure 16.2

Experiment

**Living S cells
(pathogenic
control)**

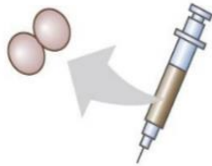


Results

Mouse dies



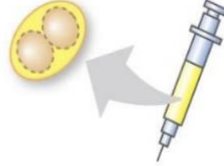
**Living R cells
(nonpathogenic
control)**



Mouse healthy



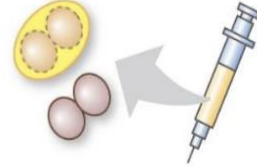
**Heat-killed S cells
(nonpathogenic
control)**



Mouse healthy



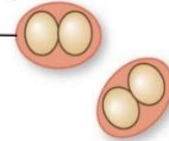
**Mixture of heat-
killed S cells and
living R cells**



Mouse dies



Living S cells



The picture shows the famous Griffith experiment, conducted in 1928, which was a fundamental step in the discovery of genetic material (DNA). The aim of the experiment was to understand how genetic traits are transmitted in the bacteria *Streptococcus pneumoniae*, which come in two forms:

- S cells (encapsulated - Pathogenic): They have an envelope that protects them from the immune system, making them pathogenic and deadly to mice.
- R cells (non-encapsulated - Nonpathogenic): They do not have a protective envelope, making them non-pathogenic and do not cause disease to mice.

Experiment details and results:

1. Injecting mice with live S cells (Pathogenic control):

- The mouse dies because the pathogenic bacteria grow and cause disease.

2. Injecting mice with live R cells (Nonpathogenic control):

- The mouse remains healthy because the non-pathogenic bacteria cannot cause disease.

3. Injecting mice with heat-killed S cells:

- The mouse remains healthy because the dead bacteria cannot reproduce or cause disease.

4. Injecting mice with a mixture of heat-killed S cells and live R cells:

- The mouse unexpectedly dies, and when its blood is examined, live S cells are found!
- This indicates that some genetic material from the dead S cells has been transferred to the live R cells, transforming them into the pathogenic S form.

Conclusion:

- The experiment showed that a genetic factor was transferred from the dead S cells to the live R cells, causing them to acquire the ability to produce the protective coating and transform into the pathogenic form.
- Later, other experiments (such as Avery's experiment in 1944) proved that this genetic factor was DNA, which led to the understanding that DNA is the

genetic material responsible for the transmission of traits.

Purpose of Genetic Transformation

- To improve the quality of crops
- To improve the productivity of crops in order to meet increasing demands of food.

Today plant biotechnology encompasses two major areas:

1. plant tissues culture
2. Plant genetic engineering.

Don by

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